for some finite truncation index $n \in \{1, 2, 3, \ldots\}$. The remainder term at order h^{n+1} will involve the integration of finitely many derivatives of its argument. Hence for observables of sufficient smoothness with compact support, this remainder term will remain bounded (see [223]).

7.7.2 First Order Operator Expansion

As we consider schemes involving compositions of three splitting pieces, the minimum number of stages to our methods (in order for it to be consistent) will be three. The perturbed Fokker-Planck operator for a splitting method with three stages will be computed as

$$\exp(hS) = \exp(hZ)\exp(hY)\exp(hX), \tag{7.18}$$

where X, Y and Z are one of $\mathcal{L}_{\mathrm{A}}^{\dagger}$, $\mathcal{L}_{\mathrm{B}}^{\dagger}$ and $\mathcal{L}_{\mathrm{O}}^{\dagger}$. Let us work the first terms of this expansion out, retaining terms involving triple products only

$$\exp(hY)\exp(hX) = \exp\left(h\left((Y+X) + \frac{h}{2}[Y,X] + \frac{h^2}{12}([Y,Y,X] + [X,X,Y]) + \ldots\right)\right),$$

where the notation $[A, B, C]$ is notation for $[A, [B, C]]$. Applying this formula one more time in (7.18), we find

$$\begin{aligned}
S &= X + Y + Z + \frac{h}{2}[Y,X] + \frac{h^2}{12}([Y,[Y,X]] + [X,[X,Y]]) \\
&\quad + \frac{h}{2}\left[Z, Y + X + \frac{h}{2}[Y,X]\right] \\
&\quad + \frac{h^2}{12}([Z,[Z,Y+X]] - [Y+X,[Z,Y+X]]) + \ldots \\
&= X + Y + Z + \frac{h}{2}([Z,Y] + [Z,X] + [Y,X]) + \frac{h^2}{4}[Z,Y,X] \\
&\quad + \frac{h^2}{12}([Y,Y,X] + [X,X,Y] + [Z,Z,Y] + [Z,Z,X] \\
&\quad + [Y+X, Y+X, Z]) + \ldots
\end{aligned}$$

In general the terms in this expansion will lead to complicated expansions. In the case of deterministic systems, the operators are always first order due to equality of mixed partials (the 2nd order derivatives are cancelled). Due to the second derivatives present in the $\mathcal{L}_{\mathrm{O}}^{\dagger}$ operator, derivatives of all orders will appear in the